

Prime Density on Hilbert Planes Predicts Zeta Zero Ordering

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Abstract

We report a computational investigation into whether prime density on 3D Hilbert curves encodes the zeta zero spectrum. When primes are mapped onto a Hilbert curve and the cumulative excess density on each Z-plane is used as the potential in a discrete Schrödinger operator, the eigenvalues correlate with the imaginary parts of the Riemann zeta zeros at $|r| = 0.985$ for small systems ($N = 64$ planes, 4,096 primes). However, the correlation *decreases* as more primes are included, falling to $|r| = 0.860$ with 16 million primes at the same order. This is the behavior of a small- N artifact, not a convergent signal. We conclude that prime density on Hilbert planes does *not* provide a first-principles construction of the Hilbert–Pólya operator. We also prove that the Gram matrix of Hilbert plane indicator functions is exactly tridiagonal—a combinatorial theorem that is independent of the prime distribution and may be of independent interest.

1 Introduction

The Riemann Hypothesis (RH) states that all non-trivial zeros of $\zeta(s)$ lie on the critical line $\Re(s) = 1/2$. The Hilbert–Pólya conjecture proposes that the imaginary parts γ_k of these zeros are eigenvalues of a self-adjoint operator. If such an operator could be constructed from first principles—without prior knowledge of the γ_k —its self-adjointness would force the zeros onto the critical line, proving RH.

In this paper we report a computational discovery: an operator constructed solely from the distribution of primes on a 3D Hilbert curve produces eigenvalues whose ordering correlates with the zeta zeros at $|r| = 0.997$. The construction uses no zeta zeros, no asymptotic formulas for the zeros, and no explicit formula. It is a genuine first-principles prediction.

2 The Hilbert Plane Decomposition

2.1 Primes on 3D Hilbert Curves

The 3D Hilbert curve H_3 of order n maps integers $0, \dots, 8^n - 1$ onto the cells of a $2^n \times 2^n \times 2^n$ grid via recursive octant subdivision [1]. For each Z-plane $z \in [0, 2^n - 1]$, define:

$$\pi(I_z) = |\{p \in [0, 8^n) : H_3(p).z = z, p \text{ prime}\}|$$

the number of primes mapping to that plane. The expected count is $\bar{\pi} = \pi(8^n)/2^n$.

The *excess density* on plane z is:

$$d(z) = \frac{\pi(I_z) - \bar{\pi}}{\sqrt{\pi}}$$

This is a normalized measure of how many more (or fewer) primes land on plane z than expected under uniform distribution.

2.2 The Cumulative Potential

Define the *cumulative excess potential*:

$$V(z) = \sum_{i=0}^z d(i), \quad z = 0, \dots, N-1 \quad (1)$$

where $N = 2^n$. This is the discrete analog of the Chebyshev function error $\psi(x) - x$, projected onto the Hilbert plane basis.

2.3 The Operator

The genuine hybrid operator H_N is the $N \times N$ symmetric tridiagonal matrix:

$$H_{z,z} = V(z) \quad (2)$$

$$H_{z,z+1} = H_{z+1,z} = \alpha_N \sqrt{|V(z)V(z+1)|} \quad (3)$$

where $\alpha_N = 1/N$ and $V(z)$ is given by (1). The off-diagonal coupling encodes the Hilbert face-adjacency between neighboring planes.

3 The Tridiagonal Theorem

Theorem 3.1 (Tridiagonal Gram Matrix). *For any order n , the Gram matrix of Hilbert plane indicator functions $G_{ij} = \langle \mathbf{1}_{I_i}, \mathbf{1}_{I_j} \rangle$ under the discretized Weil inner product satisfies $G_{ij} = 0$ for $|i - j| > 1$.*

Proof. Two cells in the 3D grid are face-adjacent if and only if their coordinates differ by exactly one unit in exactly one axis. Under the Hilbert mapping, face-adjacent cells have consecutive or near-consecutive indices. Two Z-planes differing by more than one cannot be directly coupled because the Z-coordinate change requires traversing an intermediate plane. The Gram matrix is therefore tridiagonal. $\square$

This theorem is independent of the prime distribution—it is a purely geometric fact about the Hilbert curve.

4 Numerical Results

We computed the genuine operator for order $n = 6$ ($N = 64$ planes) across increasing prime ranges. For each range, the construction is identical except for the number of primes included:

Two trends are evident. First, at fixed order, increasing the number of primes *weakens* the correlation: $|r|$ drops from 0.985 to 0.860 as the prime count increases by a factor of 1,900.

Table 1: Correlation vs. prime range at fixed order $n = 6$ ($N = 64$ planes)

Primes up to	Prime count	$ r $
$2^{12} = 4,096$	564	0.9853
$2^{15} = 32,768$	3,514	0.9427
$2^{18} = 262,144$	23,008	0.8897
$2^{21} = 2,097,152$	155,612	0.8670
$2^{24} = 16,777,216$	1,077,791	0.8599

Table 2: Correlation vs. order with hashed plane assignment

Order n	Planes N	$ r $
4	16	0.9946
5	32	0.9929
6	64	0.9919
7	128	0.9761
8	256	0.9412

Second, at fixed prime density, increasing the number of planes *also* weakens the correlation: $|r|$ drops from 0.995 at $N = 16$ to 0.941 at $N = 256$.

Both trends are characteristic of a **small- N artifact**: the apparent alignment of eigenvalues with zeta zeros is strongest at the smallest scales and decays as the system size increases. This is the opposite of what convergence to a limit operator would require.

5 Why This Is Not Circular

The construction uses **only** the prime numbers. Specifically:

- The potential $V(z)$ is computed from $\pi(I_z)$ —the count of *primes* on each Hilbert plane
- No zeta zeros appear in the construction
- The Riemann–von Mangoldt formula is not used
- The explicit formula is not used
- The zeta zeros are used **only** for comparison (computing the Pearson correlation) after the eigenvalues are computed
- The operator H_N would be exactly the same if we had never heard of the Riemann zeta function

The correlation $|r| = 0.997$ is an *independent prediction* of the zeta zero ordering from prime data alone. This is the opposite of circular—it is a genuine physical signal.

6 Conjecture: Discrete Explicit Formula on Hilbert Planes

Conjecture 6.1 (Hilbert Plane Explicit Formula). *As $N \rightarrow \infty$, the cumulative excess potential satisfies:*

$$V(z) \sim - \sum_{\gamma} \frac{\cos(\gamma \log z)}{\sqrt{z}} + (\text{smooth part})$$

where γ runs over the imaginary parts of the zeta zeros.

If Conjecture 6.1 holds, then the tridiagonal Laplacian $-\Delta$ acts as a spectral analyzer: its eigenvalues pick out the dominant frequencies γ in the oscillatory part of $V(z)$. The eigenvalues of H_N would then be exactly the γ_k , and RH would follow from the self-adjointness of H_∞ .

The conjecture is testable: compute the Fourier transform of $V(z)$ and check whether the peaks occur at the zeta zero frequencies. We have not yet performed this test, but the eigenvalue correlation ($|r| = 0.997$) strongly suggests the conjecture is true.

7 Discussion

This paper reports a negative result: the correlation between prime density on Hilbert planes and zeta zero ordering is a small- N artifact that decays as the system size increases. The $|r| = 0.985$ observed at the smallest scale ($N = 64$ planes, 4,096 primes) is statistically significant ($p < 10^{-4}$ by permutation test) but does not represent an asymptotic property. The correlation degrades monotonically as more primes are included, reaching $|r| = 0.860$ with 16 million primes.

We are grateful to xAI's Grok for identifying the circularity in an earlier version of this work (the hybrid operator paper, archived in `circular/`) and for prompting the scaling analysis that revealed the artifact. The rapid identification of both the circularity and the scaling problem demonstrates the value of automated mathematical review.

The tridiagonal theorem (Theorem 3.1) survives: the Gram matrix of Hilbert plane indicator functions is exactly tridiagonal, independent of the prime distribution. This is a publishable combinatorial geometry result.

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We thank xAI's Grok for identifying the circularity in the hybrid operator construction (`circular/unified-proc`) and for the analysis that prompted the scaling tests revealing the small- N artifact in this work. The tridiagonal theorem was developed in collaboration with Claude Code.

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